

960/480/240/120 Gbit/s Single-Carrier Super-Rated Hybrid Coherent Upstream Burst-Mode PON

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Abstract We propose and experimentally verify a hybrid coherent architecture for an upstream TDMA PON, in which multiple ONU transmitter types can be supported by a single OLT receiver. We demonstrate more than 40-dB loss budget for 480-Gbit/s using 120-GBd DP-QAM4 in upstream burst-mode reception. ©2026 The Author(s)

Introduction

Recently published ITU-T Supplement G.Supp.88 [1], which analyzed potential technical options for the impending ITU-T Very High Speed PON (VHSP) Recommendation, considers multiple transmission techniques, including intensity modulation with direct detection (IM/DD), coherent, and hybrid (combination) approaches [2] to achieve a net data throughput of 200 Gbit/s (~240 Gbit/s line rate to accommodate forward error correction (FEC) and PON protocols [1]). Historically, PONs have utilized IM/DD to minimize cost, complexity and power consumption. As data rates increase, IM/DD becomes limited by the bandwidth of optoelectronic components and by a dispersion-induced power penalty that rapidly increases with optical bandwidth. To achieve 200G in IM/DD VHSP variant, two channels of 100G, operating close to the zero-dispersion wavelength [3,4], are being considered. Coherent technology, on the other hand, offers up to fourfold increase in data rate within the same bandwidth, an improved sensitivity owing to amplification via mixing with the local oscillator, and enables digital-signal-processing-based (DSP-based) mitigation of chromatic dispersion (CD) and polarization mode dispersion (PMD) [5,6] thanks to reception of the optical field rather than optical power. However, these advantages are offset by significantly higher transceiver cost compared to IM/DD systems. To strike a balance between solution

complexity and capabilities, simplified coherent transmitters (Tx) [7-12] and receivers [10-12] are being considered. The asymmetric cost structure in PON necessitates that optical network units (ONUs) remain significantly less expensive compared to optical line terminals (OLTs), whose cost is shared across all users. Therefore, hybrid PON links with full coherent transceivers as OLTs and low-cost simplified transceivers as ONUs provides a reasonable tradeoff.

The multipoint-to-point topology of PON upstream introduces challenges for signal reception, including differential path loss of the ODN (15 dB) and the requirement to operate in burst mode to share the bandwidth using time-division multiple access (TDMA). PON-specific adaptations to the well-established coherent systems used for metro-and long-haul links are thus required [13-16]. Directly bursting the laser power for TDMA operation, as in legacy IM/DD PONs, leads to wavelength drift. To keep the Tx wavelength stable for coherent reception, an SOA booster can be used as a Tx output gate, overcoming the need for laser bursting [5]. Alternative solutions are also possible [17-19].

In this paper, we demonstrate that a single 120 GBd coherent receiver at the OLT is a universal and futureproof choice capable of supporting a variety of simplified and non-simplified upstream coherent PON schemes that

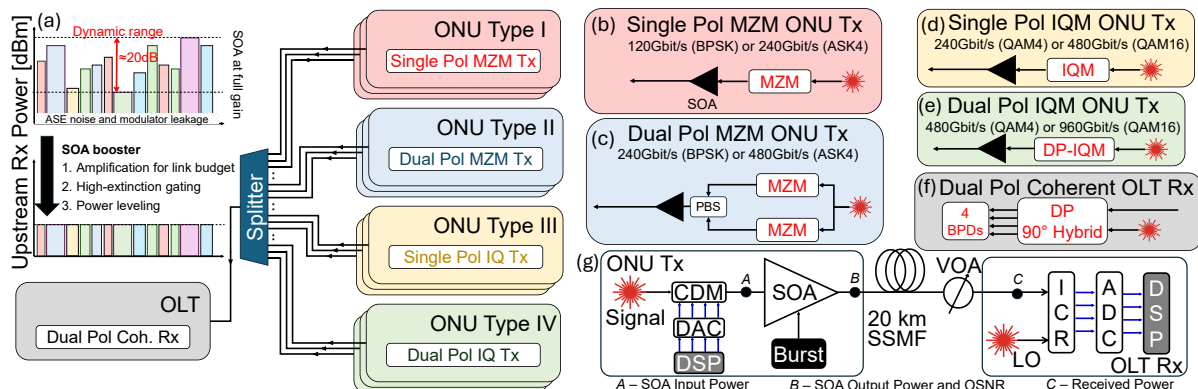


Fig. 1: (a) High-level diagram of the proposed hybrid coherent upstream architecture – SOA is used for boosting, gating and power-leveling at ONU transmitters. ONU Tx: (b) Type I – SP-MZM, (c) Type II – DP-MZM, (d) Type III – SP-IQM, (e) Type IV – DP-IQM. (f) DP-coherent OLT receiver. (g) Experimental setup used to demonstrate the proposed scheme.

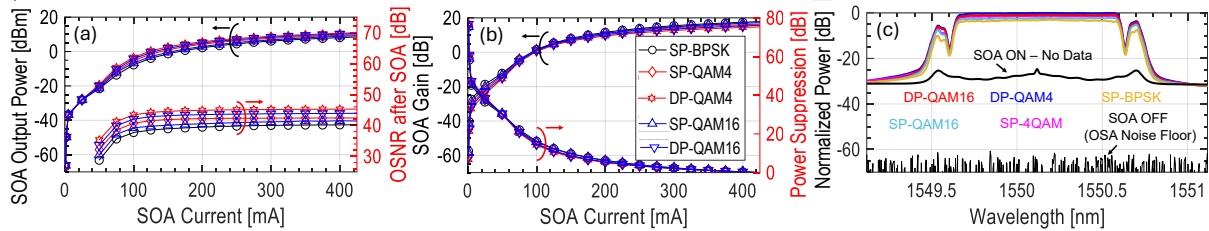


Fig. 2: (a) Booster SOA output power (left vertical axis) and output OSNR (right vertical axis). (b) Booster SOA gain (left vertical axis) and Tx power suppression (right vertical axis) as a function of booster SOA current at input powers of -9.4 , -6.5 , -3.8 , -8.0 and -5.2 dBm for SP-BPSK, SP-QAM4, DP-QAM4, SP-QAM16 and DP-QAM16, respectively. (c) Normalized spectrum at Tx output under different operating conditions.

may include at least one or more of the four different types of ONUs using conventional modulation formats and receiver DSP chain: (Type I) single-polarization (SP) Mach-Zehnder modulator (MZM), (II) dual-polarization (DP) MZM, (III) SP IQ modulator (IQM), and (IV) DP-IQM as shown in Fig.1(a-f). At the Tx, we employ an SOA operating at optimum gain to generate optical bursts. Its functionality is threefold: (i) a booster to achieve required optical power budget (OPB), (ii) a high on/off extinction ratio gate to suppress Tx emission in when not enabled (WNE) state and prevent interference accumulation to the transmitting ONU [20], and (iii) an optical power leveler to enable the use of off-the-shelf coherent receivers, not equipped with rapid TIA gain adjustment [5] on nanosecond time scales as shown in the insets of Fig.1(a).

To the best of our knowledge, we report for the first time an OPB exceeding 40 dB for a 480 Gbit/s super-rated PON operating in upstream burst mode with 120 GBd components. Additionally, we demonstrate an OPB exceeding 26 dB for 960 Gbit/s. We also show that ONUs supporting 120 Gbit/s to 480 Gbit/s with different cost and complexity could coexist in the same ODN.

Experiment

The experimental setup for demonstrating the proposed architecture is shown in Fig.1(g). The ONU Tx consists of a micro-integrable tunable laser assembly (μ TLA) operating at 1550.12 nm with an output power of 13 dBm. A coherent driver modulator (CDM) (DP-130 GBd) equipped with quad high-speed channels are driven with 120 GBd signals shaped with root-raised cosine filter with an 0.05 roll-off. Electrical signals are generated using a digital-to-analog converter (DAC) sampling at 128 GSa/s, with a 3 dB analog bandwidth exceeding 50 GHz. All the MZMs within the CDM are biased at null, typical for conventional field modulation. In the Type I Tx configuration, only one MZM is being driven, whereas in Type III, one polarization of the CDM (comprising two MZMs corresponding to in-phase and quadrature component of the same polarization) are modulated. A data rate of

120 Gbit/s is obtained using 120 GBd binary phase shift keying (BPSK) with Type I Tx. For 240 Gbit/s, single polarization (SP) quaternary quadrature amplitude modulation 4 (QAM4), also known as quaternary phase shift keying (QPSK), is employed in the Type III Tx configuration. 480 Gbit/s is realized using 120 GBd DP-QAM4 (Type IV) or SP-QAM16 (Type III). For 960 Gbit/s, Type IV Tx is used with DP-QAM16. An SOA is used to carve bursts, serving as an off gate when not enabled, and a leveling booster when enabled. Fig. 2 (c) presents the optical spectra of the various data rate signals, including cases where no signal is output from the DAC with the SOA on and off. The spectra show that without gating the SOA, there will be significant leakage from the Tx of the non-transmitting ONU that would interfere with the transmitting ONU reducing the number of ONUs that can be part of that ODN [20]. The bursts are 5 μ s long with 50% duty cycle. Fig. 3(k) shows the burst envelope measured with a low bandwidth DC-coupled receiver, whereas Fig. 3(l) shows four electrical waveforms sampled by ADC at each output of the full coherent receiver. A 20-km standard single-mode fiber (SSMF) is used to emulate the ODN along with a variable optical attenuator (VOA) that sets the required received power. At the OLT, an off-the-shelf OIF-compliant high-bandwidth DP-130 GBd micro-integrated coherent receiver (μ ICR) is used for receiving optical bursts. Another μ TLA operating close to 1550.12 nm at 15.5 dBm is used as the local oscillator (LO). The ADC is sampling at 256 GSa/s with a bandwidth of 63 GHz to digitize the signal coming from the μ ICR. For DSP, first presence of the burst is detected using DSP-based power detector. In this process, ~ 10 ns at the start of the burst is discarded to account for power detector response and Tx SOA gain response, while the last 10 ns is discarded to ensure that burst duration jitter due to the gate driver is not a problem. Subsequent DSP steps are standard and include analog front-end correction, resampling, equalization, frequency offset estimation and correction and phase recovery, after which symbols are de-mapped to bits to calculate the bit error ratio (BER).

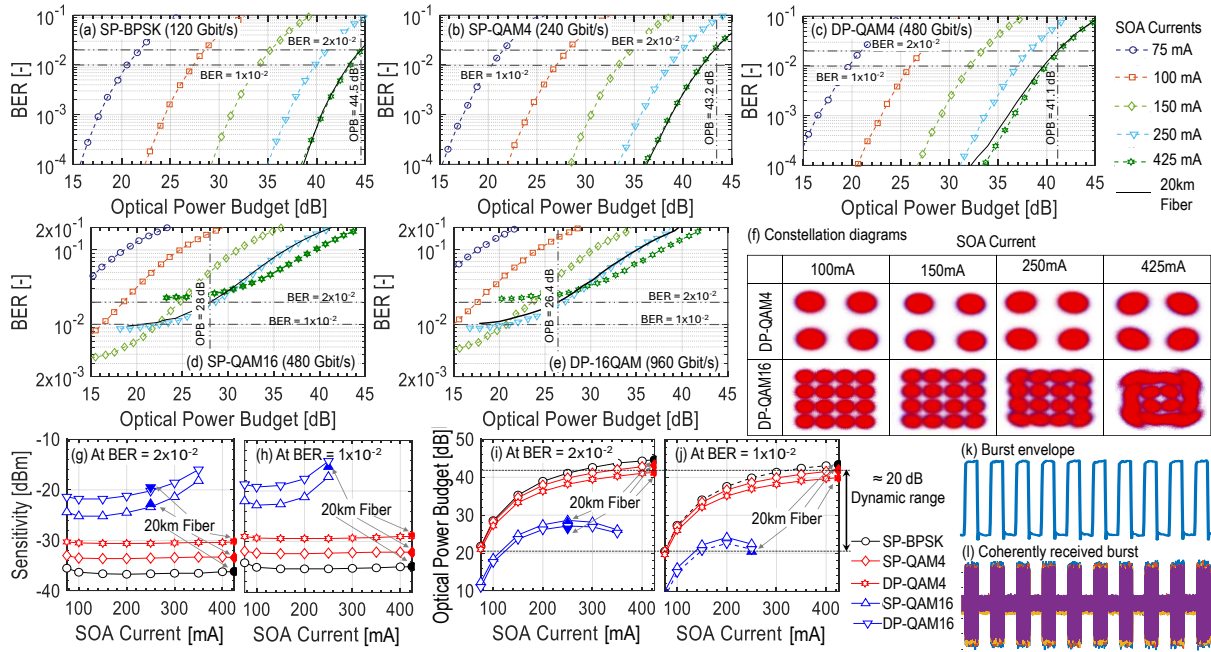


Fig. 3: BER v. optical power budget (OPB) for transmitter: (a) Type I – 120 Gbit/s SP-BPSK; (b) Type III – 240 Gbit/s SP-QAM4; (c) Type IV – 480 Gbit/s DP-QAM4; (d) Type III – 480 Gbit/s SP-QAM16; and (e) Type IV – 960 Gbit/s DP-QAM16. (f) Constellation diagrams of DP-QAM4 and DP-QAM16 signals showing progressive impact of SOA nonlinearity with increasing SOA currents. Receiver sensitivity at: (g) BER= 2×10^{-2} and (h) BER= 1×10^{-2} for different SOA currents. OPB at (i) BER= 2×10^{-2} and (j) BER= 1×10^{-2} for different SOA currents. (k) Burst envelope measured using a low bandwidth DC-coupled photodetector, and (l) burst captured measured using coherent receiver showing electrical waveforms at all four outputs for a 5 μ s burst with 50% duty cycle.

Results and Discussion

The SOA output signal power is varied by sweeping bias current and the resulting OSNR is measured and plotted in Fig. 3(a) for each modulation format. Fig. 3(b) shows the calculated SOA gain and power suppression. The output power, OSNR and gain increases with the SOA current while the power suppression decreases. The difference in the input power for different modulation formats results in different OSNRs and output powers. The transparency point for the SOA is around 100 mA, where the net gain is zero at OSNR of around 40 dB. The Tx output power can be suppressed by more than 60 dB (Fig. 2(c) SOA OFF is below the optical spectrum analyzer noise floor).

The BER is plotted against the OPB for different SOA currents in Fig. 3(a-e), for different Tx types (I-IV) and corresponding modulation formats. The fiber measurements are shown only at their optimum launch power (SOA current) (black solid line). The sensitivity and the OPB are plotted in Fig. 3(g-j) for BER = 2×10^{-2} and 1×10^{-2} , corresponding to the hard-input and soft-input threshold of the 50G-PON default LDPC (17280,14592) FEC. Though the measurements are performed applying DC bias to the SOA, no additional penalty is incurred from SOA bursting other than few ns burst ramp-up and -down time. SP signals are measured by aligning polarization of the signal at the OLT receiver using a polarization controller. A 3 dB

SNR penalty should be assumed if SP signals are received with DP coherent receiver without this alignment.

More than 40 dB OPB is achieved for data rates from 120 Gbit/s to 480 Gbit/s with a dynamic range of around 20 dB, showing the feasibility of different ONU Tx's existing in the same ODN at BER target of 2×10^{-2} . QAM16 modulation format is more susceptible to the SOA nonlinearity, limiting the maximum OPB that can be achieved for 960 Gbit/s as shown in Fig. 3(f). An OPB exceeding 26 dB can be bridged with 960 Gbit/s using DP-QAM16, however coexistence of 960 Gbit/s with lower speed ONUs under assumption of power leveling is currently precluded and requires additional counter-measures for nonlinearity (SOA with higher saturation power or DSP nonlinearity compensation).

Conclusions

We proposed and demonstrated a hybrid coherent upstream PON architecture that could simultaneously support data rates from 120 to 480 Gbit/s, enabling the use of various simplified and non-simplified ONU Tx architecture and off-the shelf coherent receiver without a rapid gain adjustment at the OLT. We use SOAs at the ONU Tx for amplifying, gating and power leveling. We achieve OPBs of more than 40 dB and 26 dB for 480 Gbit/s and 960 Gbit/s, respectively in upstream burst mode operation.

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